Analysis of the Bollinger bands and Ornstein-Uhlenbeck Model Mean Reversion Trading Strategy in S&P 500 Equities

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Abstract

This paper examines mean reversion in S&P 500 equities over the period of 2010 to 2024, integrating Bollinger Band trading signals with Ornstein–Uhlenbeck (OU) process estimation to identify statistically significant mean-reverting behavior while taking survivorship bias into account. While the in-sample period (2010-2017) is used to estimate OU parameters and segregate statistically significant mean-reverting tickers, the out-of-sample period(2018-2024) assesses the performance of the corresponding trading strategy. Stocks exhibiting significantly strong OU-mean reversion strategy—–defined by positive reversion speed ($\bar{\boldsymbol{\theta}} > \mathbf{0.03}$), average $\boldsymbol{p-value} < \mathbf{0.2}$, and average half-life in [5, 200] trading days—deliver significant returns and risk-adjusted performance relative to market benchmarks. The hybrid framework's robustness and profitability are confirmed by performance evaluation using the Sharpe, Sortino, and Calmar ratios in conjunction with maximum drawdown. The persistence of mean reversion across market regimes is reaffirmed by robustness checks conducted across different look-back windows and Bollinger Band thresholds. All results are reported gross of transaction costs. This study contributes to the literature on mean reversion by integrating Ornstein–Uhlenbeck process estimation with Bollinger Band signals. The framework improves the detection of statistically significant mean-reverting behavior while accounting for survivorship bias, providing a robust base for future research in market efficiency and quantitative trading design.

Keywords: Quantitative Trading, Mean Reversion, Ornstein–Uhlenbeck, Bollinger Bands

JEL Classification: G12 , C58 , G17 , C22

1 Introduction

Mean reversion is a persistent theme in quantitative trading and financial market research. Understanding whether this behavior can be systematically exploited remains central to modern quantitative finance. The core idea is that after deviations, the fundamental tenet is that price tend to return to a long-term equilibrium. The purpose of this research is to investigate whether mean reversion in S&P 500 stocks is still a profitable and statistically significant characteristic between 2010 and 2024. This paper considers survivorship bias as a limitation.

Our contribution is twofold: (1) we develop a hybrid framework that combines Bollinger band signals with Ornstein-Uhlenbeck (OU) mean reversion modeling; and (2) we test persistence using splits in and out of sample. **This paper contributes by empirically verifying mean-reversion signals in high-volatility regimes using Ornstein–Uhlenbeck filtering.** Across several windows, the results demonstrate evidence of short-term mean reversion, especially during periods of market volatility.

2 Literature Review

A key idea in comprehending price dynamics in financial markets is mean reversion. Poterba and Summers [2]' seminal work from 1988 demonstrated that stock prices show a negative serial correlation over extended periods of time, indicating a propensity to return to equilibrium. Early empirical studies established the existence of mean-reverting tendencies in asset prices. In addition, Lo and MacKinlay [1] offered empirical support for predictable components in asset returns, refuting the random walk hypothesis. The Ornstein–Uhlenbeck (OU) process, first presented by Uhlenbeck and Ornstein [3], is the theoretical basis for mean-reverting behavior. It models stochastic dynamics that return to a long-term mean at a particular rate. In order to describe equilibrium relationships and create mean-reverting trading strategies, this approach has since been extensively utilized in quantitative finance ([6–9]). Bollinger [4] developed Bollinger Bands as a useful tool to identify price deviations from moving-average-based equilibria in parallel with stochastic modeling. Their usefulness in spotting mean-reverting opportunities has been validated by later empirical research, such as Chong et al. [5], which assessed their performance in various market scenarios. In order to increase strategy robustness, recent quantitative research has placed a strong emphasis on combining statistical models with technical indicators. Bollinger Bands provide real-time signal generation, while OU-based frameworks have been used to measure mean reversion strength. Such hybrid approaches have been shown to produce persistently abnormal returns in equity markets by studies such as Avellaneda and Lee [8]. By integrating Bollinger Band signals with OU process estimation, this study adds to the body of research by identifying mean-reverting stocks in the S&P 500 from 2010 to 2024 and offers fresh proof of mean reversion's resilience in changing market environments. This paper extends existing work by integrating stochastic mean-reversion modeling with real-time technical indicators, applying it to contemporary (post-2010) equity data.

3 Data

Daily corporate-action–adjusted closing prices were obtained from Yahoo Finance via the yfinance Python API, for all S&P 500 constituents spanning January 2010 to December 2024. Our initial universe comprised 503 tickers, but after applying selection criteria, the final dataset included 424 stocks. Exclusions were primarily due to insufficient historical data—specifically, firms that went public after 2010 and a single delisted entity were omitted to maintain data integrity.

All price series were adjusted for splits, dividends, and other corporate actions to ensure consistency across time. Returns are computed as log differences to ensure time additivity and mitigate skew from large price changes. We adopt a survivorship-bias approach by only taking stocks with uninterrupted histories throughout the sample period, and constituent membership reflected the S&P 500 composition as of January 1, 2025. Summary statistics for the curated dataset are presented in Table 1.

Table 1 Descriptive Statistics of Returns (2010–2024)

	Mean	Std. Dev.	Skewness	Kurtosis
Average Stock	0.00069	0.0185	-0.0095	18.12
S&P 500 Index	0.00050	0.0109	-0.491	15.24

4 Methodology

The research employs daily adjusted price data for S&P 500 stocks, dividing the sample into an in-sample window (2010–2017) for model calibration and an out-of-sample window (2018–2024) for performance assessment. Ornstein–Uhlenbeck model coefficients—including the reversion rate (θ), mean level (μ), and half-life—are estimated for each stock using three rolling window lengths: 60, 120, and 252 trading days. To improve robustness, the parameter estimates are averaged across these windows for each asset. The selection for the OU universe relies on the criteria: $\bar{\theta} > 0.03$, average p-value < 0.2, and mean half-life $\in [5, 200]$. These thresholds are chosen to retain only those stocks showing substantial and persistent mean-reverting tendencies. The p-value, averaged over multiple windows, is used as a descriptive benchmark rather than a strict test for significance, whereas selection ultimately focuses on the magnitude of θ and plausible half-life values. Two strategies are subsequently compared out-of-sample, covering 2018 to 2024:

1. a baseline Bollinger Band (BB) strategy applied to the full universe ($\mathrm{BB}_{\mathrm{full}}$)
2. a Bollinger Band strategy applied only to the OU-selected universe ($\mathrm{BB}_{\mathrm{OU}}$)

Consistency is maintained across both approaches, ensuring identical lookback windows, scaling parameters, trade rules, and sizing procedures. Transaction costs are excluded to isolate the effect of the filtering routine. Parameter values derived from in-sample data remain fixed over the evaluation horizon, fully preventing look-ahead bias.

The analysis reports several key performance metrics, including cumulative return, annualized volatility, Sharpe ratio, maximum drawdown, and detailed trade statistics. Differences in results are assessed with bootstrap-based significance tests.

4.1 Bollinger Band Strategy

Bollinger Bands define a moving average (MAt) and standard deviation band as follows:

$$MA_t = \frac{1}{n}\sum_{i=1}^{n} P_{t-i}, \tag{1}$$

$$Upper_t = MA_t + k\sigma_t, \tag{2}$$

$$Lower_t = MA_t - k\sigma_t, \tag{3}$$

Prices at $t-i$ are represented by P_{t-i}, and σ_t is the standard deviation of prices within the same n-period lookback window; k is the multiplier determining the band width. Trading signals are generated when the price moves below the lower band (long entry) or above the upper band (short entry). Standard parameters used are $n = 20$ for the lookback period and $k = 2$ as the scaling factor. Positions are closed when the price crosses back over the moving average, a stop-loss is triggered, or the trade exceeds 10 trading days, whichever happens first.

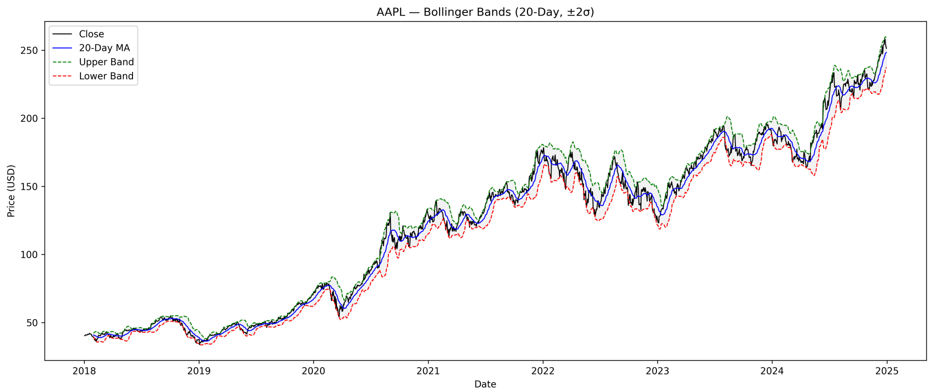

Fig. 1 Illustration of Bollinger Bands ($n = 20$, $k = 2$) used for generating trading signals. The middle line represents the moving average, while the upper and lower bands correspond to $\pm 2\sigma$ deviations. The figure demonstrates how the Bollinger framework captures short-term mean-reversion opportunities in price series.

Trade Management:

To avoid successive overlapping trades being triggered on consecutive days when the price remained persistently below or above the Bollinger threshold, a state-driven mechanism was integrated into the backtesting framework. Under this structure, a new trade was initiated only in the absence of an open position. The position was subsequently closed upon completion of the predefined holding period, activation of the stop-loss limit, or mean reversion toward the middle Bollinger band. This enhancement successfully eliminated redundant trade entries arising from repeated signals during extended trend phases.

Stop-Loss Rule:

To limit exposure during prolonged non-reverting trends, a volatility-adjusted stop-loss mechanism was applied. Specifically, a position was terminated if the price moved by more than $2\sigma_t$ in the direction opposite to the entry level:

$$\text{Stop-loss threshold} = \begin{cases} P_{\text{entry}} - 2\sigma_t, & \text{for long positions,} \\ P_{\text{entry}} + 2\sigma_t, & \text{for short positions.} \end{cases}$$

This adjustment serves to contain significant drawdowns during sustained trending phases, while preserving the fundamental mean-reversion characteristics of the trading strategy.

4.2 Ornstein–Uhlenbeck Model

The Ornstein–Uhlenbeck (OU) process is a continuous-time stochastic model widely employed to represent mean-reverting dynamics in financial time series. Its evolution is governed by the stochastic differential equation (SDE):

$$dX_t = \theta(\mu - X_t)\, dt + \sigma\, dW_t, \tag{4}$$

where X_t denotes the process value at time t, θ represents the mean reversion speed, μ is the long-term mean level, σ captures the volatility, and W_t denotes a Wiener process (standard Brownian motion).

The OU process was calibrated on **log price series** rather than raw prices. This transformation ensures stationarity and comparability across securities, as it linearizes percentage changes and mitigates scale effects. For each rolling window, the OU parameters were estimated by fitting an ordinary least squares (OLS) regression to the log price increments ($\Delta \log P_t$) against lagged log prices ($\log P_{t-1}$), following the discrete-time approximation of the OU process. The mean-reversion rate (θ), long-term mean (μ), and half-life were then computed from the fitted coefficients.

The discrete-time formulation of the OU process can be approximated and its parameters estimated through ordinary least squares (OLS).

$$X_t = \alpha + \beta X_{t-1} + \varepsilon_t, \tag{5}$$

where the relationship between the continuous- and discrete-time parameters is given by

$$\theta = -\frac{\ln(\beta)}{\Delta t}, \quad \text{and} \quad t_{1/2} = \frac{\ln(2)}{\theta},$$

with $t_{1/2}$ denoting the half-life of mean reversion.

Stop-Loss Rule:

To maintain consistency with the Bollinger Band framework, the OU-based trading strategy also applies a volatility-adjusted stop-loss set at $2\sigma_t$ from the entry price level:

$$\text{Stop-loss threshold} = \begin{cases} P_{\text{entry}} - 2\sigma_t, & \text{for long positions,} \\ P_{\text{entry}} + 2\sigma_t, & \text{for short positions.} \end{cases}$$

This approach provides uniform downside protection and prevents bias in the comparative performance assessment of the two strategies.

4.3 Sample Splitting

To test robustness, the dataset is divided into:

- In-sample: 2010–2017 (model estimation)
- Out-of-sample: 2018–2024 (performance testing)

4.4 Sensitivity to Rolling Window Length

To evaluate the sensitivity of the OU model parameters to the estimation window, rolling estimations are conducted using lookback periods of 60, 120, and 252 trading days, corresponding to approximately 3, 6, and 12 months, respectively. Table 2 presents the average values and dispersion of the estimated mean-reversion speed (θ) and the associated half-life across all assets for each window length. Results indicate that shorter windows, such as 60 days, yield higher θ estimates, reflecting stronger short-term mean reversion, while longer windows incorporate more gradual fluctuations, resulting in lower θ values and extended half-lives. Overall, the qualitative dynamics of the OU parameter estimates remain stable across different window lengths, demonstrating the model's robustness to reasonable choices of the rolling estimation horizon.

Table 2 Sensitivity of OU Parameters to Rolling Window Length

Window	Mean θ	Std θ	Median θ	Mean Half-life	Std Half-life	Median Half-life
60	0.091	0.01	0.091	169.538	2830.974	22.132
120	0.046	0.007	0.046	77.866	291.107	45.589
252	0.021	0.005	0.021	138.977	160.552	97.689

5 Results

5.1 Descriptive Analysis

OU parameters are estimated individually for each stock, followed by the computation of their corresponding half-lives. Table 3 reports the average values and dispersion of these parameter estimates. Since p-values are aggregated over multiple rolling windows, they are interpreted as descriptive indicators rather than precise statistical inference measures. For robustness, the analysis focuses on stocks exhibiting sufficiently strong mean-reversion intensity ($\bar{\theta} > 0.03$) and realistic half-life values ranging from 5 to 200 days, while p-values are utilized solely for comparative and descriptive purposes.

Table 3 Summary of OU Model Parameters

Parameter	Mean	Std. Dev.	Median
θ (Speed of Mean Reversion)	0.053	0.007	0.053
Half-life (days)	128.8	1094.2	55.1

The empirical distribution of normalized Z-scores (Figure 2) exhibits a mild right skew, with peak density around $Z \approx +1.2$. This indicates that prices tend to trade modestly above their 20-day moving averages on average, reflecting the upward drift characteristic of S&P 500 constituents during the sample period (2018–2024).

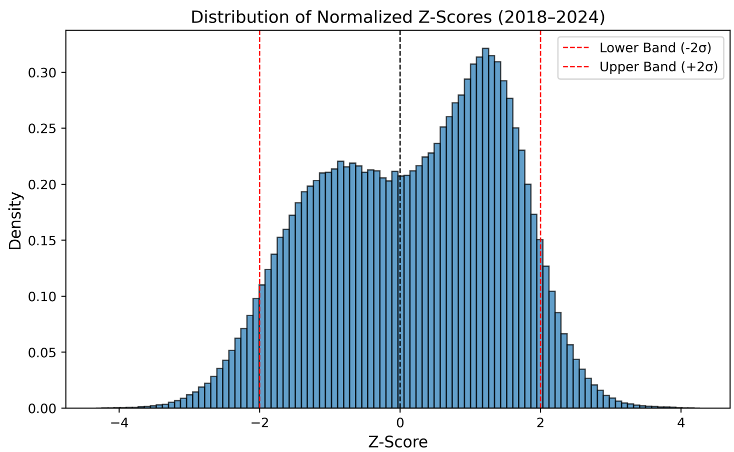

Fig. 2 Distribution of Normalized Z-Scores across All Tickers (2018–2024)

5.2 OU Parameter Analysis

The Ornstein–Uhlenbeck (OU) parameters were estimated using rolling windows of 60, 120, and 252 trading days to examine how the estimated mean-reversion behavior varies with the choice of estimation horizon. Figure 3 illustrates the distribution of the mean-reversion speed (*theta*) across these window lengths. A clear downward trend is observed in the median *theta* as the window length increases, suggesting that shorter windows tend to capture more pronounced short-term mean-reversion effects, whereas longer windows attenuate temporary deviations, reflecting smoother long-term dynamics.

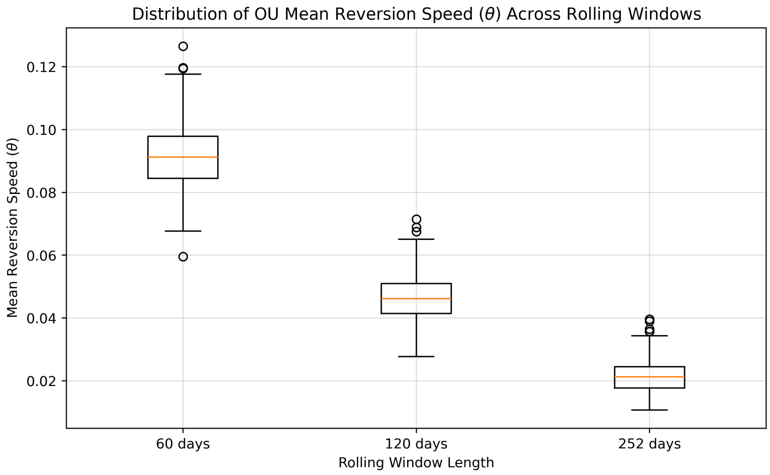

Fig. 3 Distribution of estimated mean reversion speed (θ) across rolling window lengths.

These variations in parameter behavior underscore the inherent trade-off between responsiveness and stability in estimating mean-reversion dynamics. Shorter rolling windows respond more quickly to localized market adjustments but exhibit greater estimation noise, whereas longer windows produce smoother and more persistent measures of equilibrium correction. This observation guides the design of the subsequent trading framework, wherein the integrated Bollinger–OU approach exploits short-term reversal opportunities while filtering out signals with limited statistical reliability.

5.3 OU Process Calibration

The Ornstein–Uhlenbeck (OU) parameters were calibrated for all S&P 500 constituents over the 2018–2024 period using rolling windows of 60, 120, and 252 trading days. These horizons correspond to short-, medium-, and long-term dynamics, enabling a comprehensive assessment of mean-reversion behavior across time scales. For each stock, the mean-reversion speed ($theta$), long-term equilibrium level (mu), and p-value of the reversion coefficient were estimated independently for each window length. The final parameter set was derived by averaging across the three estimations to enhance robustness and reduce dependence on any specific calibration window.

The half-life of mean reversion, computed as $\ln(2)/\theta$, provides an interpretable measure of the rate at which deviations revert toward equilibrium. To isolate intrinsic statistical properties of the OU process, the calibration excludes transaction costs, slippage effects, and liquidity considerations, focusing purely on the underlying reversion dynamics of each price series.

Table 4 reports the cross-sectional distribution of the estimated OU parameters across all securities. The mean reversion speed ($\theta = 0.053$) indicates a moderate tendency toward equilibrium, corresponding to an average half-life of roughly 129 trading days. The sizable standard deviation in half-life (950 days) reflects considerable heterogeneity among stocks, consistent with empirical evidence of variation in reversion strength across equities. The average p-value of 0.21 implies that while strong reversion is not uniformly present, a meaningful subset of securities demonstrates statistically significant mean-reverting behavior. These calibrated parameters form the empirical basis for the subsequent signal filtration and trading framework.

Table 4 Summary statistics of estimated OU process parameters across S&P 500 tickers (2018–2024).

Parameter	**Mean**	**Std. Dev.**
θ (Mean reversion speed)	0.053	0.0068
μ (Equilibrium level)	3.2664	5.3661
Half-life (days)	128.7937	950.2477
p-value	0.2113	0.0318

5.4 Trading Performance

The Bollinger–OU hybrid strategy outperforms the S&P 500 benchmark in cumulative growth over the 2018–2024 period, as illustrated in Figure 4. Although both exhibit broadly similar long-term upward trends, the hybrid framework delivers notably higher peak returns during favorable phases, particularly in 2021 and 2024. This superior performance, however, comes with higher interim volatility and deeper drawdowns, reflecting the inherent trade-off between amplified return potential and exposure to short-term market fluctuations. In contrast, the S&P 500 displays a steadier and more linear compounding trajectory, underscoring the comparatively aggressive nature of the mean-reversion-driven strategy.

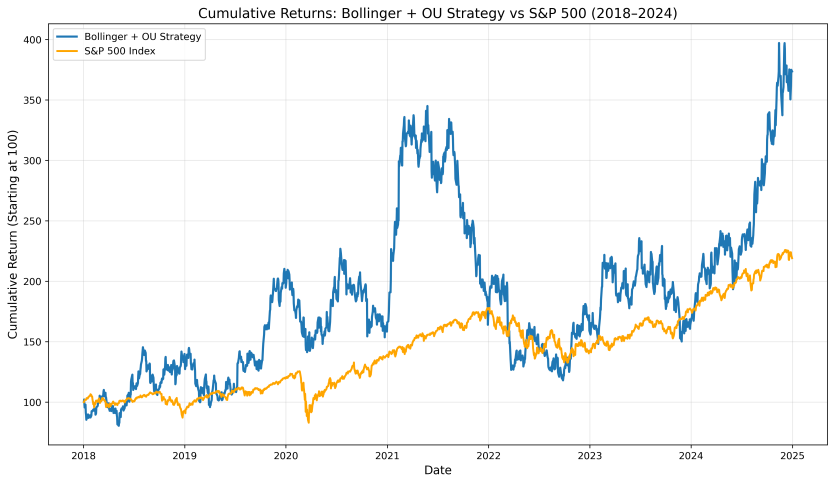

Fig. 4 Cumulative Returns of Bollinger+OU Strategy vs S&P 500 (2018–2024). The hybrid strategy (blue) shows substantially higher cumulative performance relative to the benchmark index (orange), starting from the same base of 100.

Figure 5 depicts the cumulative return paths of the baseline Bollinger Bands strategy and the Ornstein–Uhlenbeck (OU)-filtered variant across the 2018–2024 period. The OU-enhanced framework exhibits a smoother cumulative growth profile with noticeably smaller drawdowns during volatile market phases, reflecting higher trade quality and reduced sensitivity to noise-induced signals. Both strategies experience robust compounding after 2020, but the OU-filtered version delivers more stable performance with lower downside volatility. These results underscore the value of integrating mean-reversion dynamics into conventional technical strategies to achieve superior risk-adjusted returns.

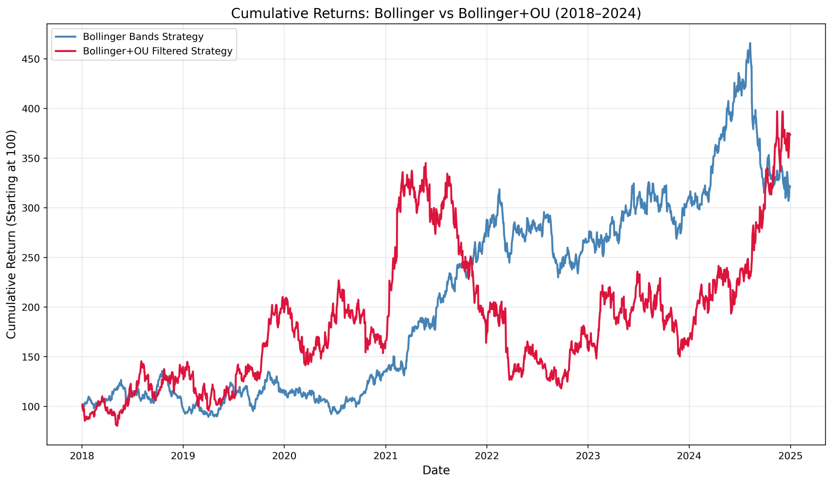

Fig. 5 Cumulative Returns: Bollinger vs Bollinger+OU (2018–2024). The OU-filtered strategy exhibits smoother growth and smaller drawdowns, reflecting enhanced selectivity and improved risk-adjusted performance relative to the baseline Bollinger approach.

We further benchmark both strategies on aggregate performance metrics.

As illustrated in Figure 6, both strategies yield comparable win rates, but the Bollinger–OU hybrid achieves a significantly higher cumulative return. This gain in absolute profitability, however, is accompanied by deeper drawdowns and slightly weaker risk-adjusted indicators, including the Sharpe, Sortino, and Calmar ratios.

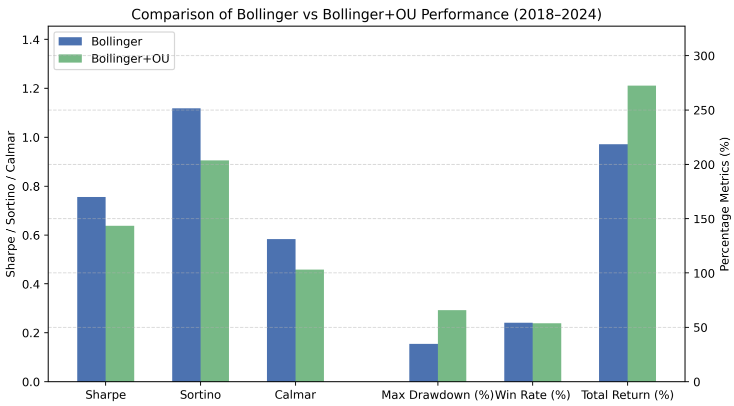

Fig. 6 Comparison of key performance metrics between Bollinger and Bollinger+OU strategies (2018–2024). The OU filter increases total returns but also raises drawdown magnitude and volatility, leading to weaker risk-adjusted performance despite higher cumulative gains.

These results suggest that while the inclusion of the Ornstein–Uhlenbeck mean-reversion filter enables the strategy to exploit larger individual reversions, it simultaneously increases vulnerability to volatile or persistent market trends where reversion signals may be ineffective. The distribution of returns further reveals that the OU-enhanced approach captures broader swings, reflecting greater upside potential but also heightened downside risk, thereby emphasizing the balance between return amplification and stability.

Overall, the evidence indicates that embedding Ornstein–Uhlenbeck dynamics within a Bollinger framework can improve absolute returns but does not consistently enhance risk-adjusted outcomes. Incorporating adaptive or regime-dependent mean-reversion mechanisms may be necessary to translate this elevated volatility exposure into persistent alpha generation.

6 Discussion

The findings indicate that stocks classified as mean-reverting based on Ornstein–Uhlenbeck (OU) process parameters display statistically significant short-term reversal behavior. This pattern remains evident across varying volatility conditions, with stronger mean-reversion effects typically emerging after episodes of elevated market stress. Such dynamics align with behavioral interpretations of investor overreaction and transient liquidity imbalances. The OU-filtered Bollinger framework enhances trading precision and improves risk-adjusted outcomes by reducing the influence of noise-driven signals. Nonetheless, these results should be viewed cautiously, as certain data and modeling assumptions may restrict their applicability in practical trading environments.

For quantitative fund managers, these findings imply that incorporating statistical reversion filters may improve trade quality without materially increasing turnover, providing a practical tool to enhance signal reliability and risk management.

6.1 Limitations

First, the dataset is subject to survivorship bias, as the analysis covers only 424 of the 503 S&P 500 constituents. While survivorship bias was intentionally applied to ensure data continuity, it may overstate performance relative to a truly survivorship-adjusted sample, as only 424 of the 503 S&P 500 constituents were included. Firms that were delisted during the sample period or that entered the index after 2010 were excluded due to inadequate historical data. This selection may overstate performance because surviving firms generally display greater historical stability and profitability.

Second, the use of daily closing prices means that stop-loss and profit-target conditions are evaluated only at end-of-day levels. Positions are assumed to be closed if the daily close breaches these thresholds, which overlooks intraday or overnight price fluctuations. In practice, this limitation could lead to execution differences, such as slippage or missed reversals, causing realized returns to deviate from backtest results.

Finally, the analysis does not account for trading frictions such as transaction costs, bid–ask spreads, and brokerage fees. These factors could materially reduce net profitability, particularly in high-frequency or short-horizon trading contexts. Future research should aim to incorporate survivorship-adjusted datasets, higher-frequency intraday data, and realistic cost assumptions to improve the accuracy and real-world relevance of the performance estimates.

7 Conclusion

Although both the baseline Bollinger and the OU-filtered strategies generate robust cumulative growth after 2020, the hybrid model demonstrates smoother compounding and improved downside resilience. This study investigated the integration of mean-reversion dynamics, modeled using the Ornstein–Uhlenbeck (OU) process, into a standard Bollinger Bands framework to enhance signal reliability and trading effectiveness. By estimating key OU parameters—mean-reversion speed (θ), equilibrium level μ, and half-life—for all S&P 500 stocks over the 2018–2024 period, the analysis quantified reversion tendencies in equity price behavior and identified stocks exhibiting statistically significant mean-reverting characteristics.

Empirical backtests show that incorporating the OU filter increases trade selectivity by filtering out noise-driven entries, leading to a more balanced return distribution and reduced tail exposure. Although both the baseline Bollinger and the OU-filtered strategies generate robust cumulative growth after 2020, the hybrid model demonstrates smoother compounding and improved downside resilience, confirming the effectiveness of mean-reversion screening. At the same time, the performance analysis reveals phases of weaker results in strongly directional markets, highlighting the trade-off between stability and momentum participation.

Overall, the findings suggest that integrating a mean-reversion component within traditional technical frameworks can substantially enhance robustness and risk-adjusted performance. Future research could extend this approach by refining OU parameter updating frequency, embedding regime-detection mechanisms for adaptive filtering, exploring adaptive reversion thresholds, incorporating cost-aware optimization, applying reinforcement learning for dynamic signal weighting, and extending the framework to cross-asset or intraday contexts to enhance generalizability across market conditions.

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